



NATIONAL UNIVERSITY - MANILA
COLLEGE OF COMPUTING AND INFORMATION TECHNOLOGY

Introduction to Machine Learning

LOGISTIC REGRESSION

(PREDICTING CUSTOMER PURCHASE)

ASSIGNMENT NO. 3

Prepared by:

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The (*) symbols means that the answer is rounded up.

Dataset

Customer	Time on Site (X1)	Pages Viewed (X2)	Purchase (y)
A	1	4	0
B	2	3	0
C	3	7	1
D	5	2	1
E	6	6	1

Initial Model Parameters:

- $m_1 = 0.8$
- $m_2 = 0.4$
- $b = -4.0$

1. Compute Probabilities (5 Points)

For each customer A to E:

- Compute the linear score
- Compute the predicted probability

Customer	Time on Site (X1)	Pages Viewed (X2)	Purchase (y)	$\hat{y}$
A	1	4	0	0.168*
B	2	3	0	0.231
C	3	7	1	0.769*
D	5	2	1	0.690*
E	6	6	1	0.961*

The (*) symbols means that the answer is rounded up.

Solution:

LINEAR SCORE:

$$Z = w_1 x_1 + w_2 x_2 + b$$

PREDICTED PROBABILITY:

$$\hat{y} = \sigma(z)$$

$A = 0.8(1) + 0.4(4) + (-4)$
 $= \underline{-1.6}$

$B = 0.8(2) + 0.4(3) + (-4)$
 $= \underline{-1.2}$

$C = 0.8(3) + 0.4(7) + (-4)$
 $= \underline{1.2}$

$D = 0.8(5) + 0.4(2) + (-4)$
 $= \underline{0.8}$

$E = 0.8(6) + 0.4(6) + (-4)$
 $= \underline{3.2}$

$A = \frac{1}{1 + e^{-(-1.6)}} = \underline{0.168^*}$

$B = \frac{1}{1 + e^{-(-1.2)}} = \underline{0.231}$

$C = \frac{1}{1 + e^{-(1.2)}} = \underline{0.769^*}$

$D = \frac{1}{1 + e^{-(0.8)}} = \underline{0.690}$

$E = \frac{1}{1 + e^{-(3.2)}} = \underline{0.961^*}$

2. Compute The Average Loss (6 Points)

For each customer A to E:

- Compute the Binary Cross Entropy loss for each sample

Customer	Time on Site (X1)	Pages Viewed (X2)	Purchase (y)	$\hat{y}$	Loss
A	1	4	0	0.168*	0.1839
B	2	3	0	0.231	-0.2627*
C	3	7	1	0.769*	0.2627*
D	5	2	1	0.690*	0.3711*
E	6	6	1	0.961*	0.0398*

Average Loss	0.2240*
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The (*) symbols means that the answer is rounded up.

Solution:

COMPUTE THE AVERAGE LOSS

$$\text{LOSS} = -(y_i \times \ln(\hat{y}_i) + (1 - y_i) \times \ln(1 - \hat{y}_i))$$

Sum of Loss = 1.1202

$$A = -(0 - \ln(0.168) + (1 - 0) \times \ln(1 - 0.168))$$

$$= 0.1839$$

$$B = -(0 - \ln(0.231) + (1 - 0) \times \ln(1 - 0.231))$$

$$= 0.2626 \approx 0.2627$$

$$C = -(1 - \ln(0.769) + (1 - 1) \times \ln(1 - 0.769))$$

$$= 0.2626 \approx 0.2627$$

$$D = -(1 - \ln(0.690) + (1 - 1) \times \ln(1 - 0.690))$$

$$= 0.3710 \approx 0.3711$$

$$E = -(1 - \ln(0.961) + (1 - 1) \times \ln(1 - 0.961))$$

$$= -0.0397 \approx 0.0398$$

$$AL = \frac{1.1202}{5} = 0.2240$$

3. Update the slope and intercept using Gradient Descent (6 Points)

Perform one step of gradient descent with learning rate $\eta = 0.1$:

Customer	Time on Site (X1)	Pages Viewed (X2)	Purchase (y)	$\hat{y}$	$\hat{y} - y$
A	1	4	0	0.168*	0.168
B	2	3	0	0.231	0.231
C	3	7	1	0.769*	-0.231
D	5	2	1	0.690*	-0.310
E	6	6	1	0.961*	-0.039

Compute gradients of the loss w.r.t. m_1 , m_2 , b

$\partial L / \partial m_1$	-0.3694
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$\partial L / \partial m_2$	-0.2212
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$\partial L / \partial b$	-0.0362
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Calculate the new slopes m_1 , m_2 and intercept b :

new m_1	0.8369
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new m_2	0.4221
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new b	-3.996
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The (*) symbols means that the answer is rounded up.

Solution:

m_1

$$A = 0.168 \times 1 = 0.168$$

$$B = 0.231 \times 2 = 0.462$$

$$C = -0.231 \times 3 = -0.693$$

$$D = -0.310 \times 5 = -1.550$$

$$E = -0.039 \times 6 = -0.234$$

$$\underline{-1.847}$$

$$\frac{\partial L}{\partial m_1} = \frac{-1.847}{5} = \underline{\underline{-0.3694}}$$

m_2

$$A = 0.168 \times 4 = 0.672$$

$$B = 0.231 \times 3 = 0.693$$

$$C = -0.231 \times 7 = -1.617$$

$$D = -0.310 \times 2 = -0.620$$

$$E = -0.039 \times 6 = -0.234$$

$$\underline{-1.106}$$

$$\frac{\partial L}{\partial m_2} = \frac{-1.106}{5} = \underline{\underline{-0.2212}}$$

b

$$\Sigma(\hat{y} - y) = -0.181$$

$$\frac{\partial L}{\partial b} = \frac{-0.181}{5} = \underline{\underline{-0.0362}}$$

The (*) symbols means that the answer is rounded up.

CALCULATE THE NEW SLOPES

$$\begin{aligned} \text{new } m_1 &= m_1 - \eta \frac{\partial L}{\partial m_1} \\ &= 0.8 - 0.1 \frac{-1.847}{5} \\ &= 0.8 - (-0.03694) \\ &= \underline{\underline{0.8369}} \end{aligned}$$

$$\begin{aligned} \text{new } m_2 &= m_2 - \eta \frac{\partial L}{\partial m_2} \\ &= 0.4 - 0.1 \frac{-1.106}{5} \\ &= 0.4 - (-0.02212) \\ &= \underline{\underline{0.4221}} \end{aligned}$$

$$\begin{aligned} \text{new } b &= b - \eta \frac{\partial L}{\partial b} \\ &= -4.0 - 0.1 \frac{-0.181}{5} \\ &= -4.0 - (-0.0036) \\ &= \underline{\underline{-3.996}} \end{aligned}$$

4. Compute new probabilities using the new slope and intercept (5 Points)

Customer	Time on Site (X1)	Pages Viewed (X2)	Purchase (y)	new $\hat{y}$
A	1	4	0	0.187*
B	2	3	0	0.258
C	3	7	1	0.813*
D	5	2	1	0.737
E	6	6	1	0.972

The (*) symbols means that the answer is rounded up.

Solution:

LINEAR SCORE

$$A = 0.8369(1) + (0.4221)(4) + (-3.996) = -1.4707$$

$$B = 0.8369(2) + (0.4221)(3) + (-3.996) = -1.0559$$

$$C = 0.8369(3) + (0.4221)(7) + (-3.996) = 1.4694$$

$$D = 0.8369(5) + (0.4221)(2) + (-3.996) = 1.0327$$

$$E = 0.8369(6) + (0.4221)(6) + (-3.996) = 3.558$$

PREDICTED PROBABILITY

$$A = 1 / (1 + e^{-(-1.4707)}) = 0.1868 \approx 0.187$$

$$B = 1 / (1 + e^{-(-1.0559)}) = 0.2580$$

$$C = 1 / (1 + e^{-(1.4694)}) = 0.81296 \approx 0.8130$$

$$D = 1 / (1 + e^{-(1.0327)}) = 0.7374$$

$$E = 1 / (1 + e^{-(3.558)}) = 0.97229 \approx 0.9723$$

5. Compute new Average Loss (6 Points)

Customer	Time on Site (X1)	Pages Viewed (X2)	Purchase (y)	new $\hat{y}$	new Loss
A	1	4	0	0.187*	0.2070
B	2	3	0	0.258	0.2984
C	3	7	1	0.813*	0.2070
D	5	2	1	0.737	0.3052*
E	6	6	1	0.972	0.0284*

Average Loss	0.2092
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The (*) symbols means that the answer is rounded up.

Solution:

COMPUTE THE NEW AVERAGE LOSS

$$\begin{aligned} A &= -(0 \times \ln(0.187) + (1-0) \times \ln(1-0.187)) = 0.2070 \\ B &= -(0 \times \ln(0.258) + (1-0) \times \ln(1-0.258)) = 0.2984 \\ C &= -(1 \times \ln(0.813) + (1-1) \times \ln(1-0.813)) = 0.2070 \\ D &= -(1 \times \ln(0.737) + (1-1) \times \ln(1-0.737)) = 0.3051 \approx 0.3052 \\ E &= -(1 \times \ln(0.972) + (1-1) \times \ln(1-0.972)) = 0.0283 \approx 0.0284 \end{aligned}$$

$$\text{SUM OF LOSSES} = 1.046$$

$$= 1.046 / 5$$

$$AL = \underline{\underline{0.2092}}$$